

# Learning Football through Interactive Storytelling

David Xu Hu, Yuning Yao

<sup>1</sup> LIACS, Leiden University, Niels Bohrweg 1, Leiden, The Netherlands  
[y.yao.7@umail.leidenuniv.nl](mailto:y.yao.7@umail.leidenuniv.nl), [d.xu.hu@umail.leidenuniv.nl](mailto:d.xu.hu@umail.leidenuniv.nl)

supervised by Coen Werre, Fons Verbeek  
[c.p.d.werre@fsw.leidenuniv.nl](mailto:c.p.d.werre@fsw.leidenuniv.nl),  
[f.j.verbeek@liacs.leidenuniv.nl](mailto:f.j.verbeek@liacs.leidenuniv.nl)

**Abstract.** This paper presents an innovative interactive storytelling web platform aimed at teaching football rules and concepts to individuals unfamiliar with the sport but keen on books and games. The problem addressed is the lack of accessible and engaging educational tools for understanding football, which can be intimidating for newcomers. The presented solution, inspired by book-like interfaces and library navigation experiences, combines them with interactive simulations and mini-games to offer an immersive learning environment. This approach enhances user engagement through appealing design and intuitive functionality, making complex rules more understandable and more enjoyable to learn. The results show that the website’s creative approach and interactions offer a better alternative to traditional learning resources.

## 1 Introduction

### 1.1 General Background and Problem Definition

Football is the most popular sport in the world (WorldAtlas 2023). It is also a complex sport, with many small nuances and intricacies that make it hard to follow beyond people kicking a ball (O’Hanlon 2022). Although there is no scientific research on the topic of how easy it is for a beginner to learn about football, there are bound to be those who feel daunted by its many rules and concepts.

Multiple studies have shown that having learners actively involved can significantly improve the learning process. However, resources freely available online are often in the form of basic websites using text and pictures (US National Soccer Players n.d.) (Soccer.com 2022) (World Soccer Talk n.d.). There are grounds to believe that this lack of interaction and creative information visualizations is a significant part of the reason why some beginners find it hard to learn more about the game.

Consequently, our project aims to fill that gap, having the following research question:

- Can we use interactive information visualizations and a book-like experience to engage users in learning football rules and dynamics in a more enjoyable way?

## 1.2 Approach: Bridging Knowledge Gaps Through Interactive Learning

Besides the different creative interactions, to give users a sense of familiarity, the project uses the idea of books to tie all of the different interactive information visualizations together. It includes 3 main components:

- **Introductory Book:** Story about and introducing football presented like a book, simulating the familiarity of traditional learning mediums.
- **Interactive Simulations and Mini-Games:** Engaging the user in creative interactive experiences to enhance learning and information retention
- **Library Hub:** Offering interactive, structured access to other football-related topics.

The design draws from theories of experiential learning (Kolb 2014) and interactive information visualization, incorporating principles like usability, visibility, and task orientation to ensure a user-friendly experience. The usability goals for the project are the following:

- **Effectiveness:** Helping users learn and apply basic football rules.
- **Clarity:** Ensuring the product is easily understandable and intuitive to use
- **Satisfaction:** Delivering a visually appealing and interactive experience.

About the intended audience of the product, the focus is on individuals aged 18–35 with minimal exposure to football exposure, looking to learn more about it and keen on books and games. These users prefer visually rich, structured, and intuitive platforms that simplify complex concepts without oversimplification. Existing solutions often fail to meet these needs, leaving a gap for more immersive and effective approaches.

## 1.3 Paper Outline

This paper begins by defining the background and relevance of the problem. It then describes the solution, detailing its design rationale and usability requirements. Following this, the Materials & Methods section, which outlines the tools and technologies used in development. Finally, there are the results derived from the evaluations done, and a discussion of key insights and challenges, as well as suggestions for future work.

## 2 Material & Methods

The project is web-based, using a combination of HTML, CSS, and JavaScript. The three.js library is used for the interactive 3D elements, and the d3.js library is used for creative 2D visualizations. The addition of the three.js and d3.js libraries did not come until some time into the project, as we found them while looking for ideas on how to make interesting interactions and visualizations. The website is supposed to be used only in a computer, as it is not adapted for mobile phones or other platforms. Initial designs and prototypes were created using tools like Procreate and Figma for interface design and wireframing, ensuring a user-friendly layout and aesthetic.

### Methodology:

- **Design and Development:** A user-centered design approach was adopted, focusing on iterative prototyping and feedback. The process began with wireframes and low-fidelity prototypes, followed by the integration of interactive elements and visualizations.
- **Platform Structure:** The project is divided into two main components: A structured, interactive storytelling module designed to teach basic football concepts; Three types of library offering deeper dives into football topics.
- **Evaluation:** The prototype was tested for usability and effectiveness. User feedback was collected to identify and address pain points.

## 3 Results and Evaluation

The two user evaluations that were performed played a key role in guiding the development process, helping refine the platform to ensure it offers a functional and intuitive experience. The system received generally positive feedback regarding its usability, interactivity, and educational potential, but certain limitations were identified, particularly in navigation, animations, and the visibility of key interactive elements. For a look into the final prototype, refer to the appendix.

For the first evaluation, the focus was on having the functionalities of the introductory book ready for the prototype, so it included all functionalities of the book itself, the formations simulation, the field map simulation, and definitions notebook. 4 users were gathered for the test, with limited football knowledge, but an interest in learning through books and interactive experiences.

The results for it were very good, with an average SUS score of 92.5. All evaluation metric goals were met, and usability factors were good. The most important source of feedback were the suggestions given by the participants. Some users thought the icons of the navigation bar would be better with additional visibility. They also mentioned some possible adjustments to the simulations, and lastly the information content was

thought to be a possible point of improvement. Notably some users thought the perceived affordances of the notebook were not very clear, and had a hard time going back through the layers. All doable suggestions were taken into consideration for further improvement of the website, barring the ones related to content, as the focus was still only on the platform's functionalities and interactions.

Next came user evaluation 2. This evaluation's prototype included the library hub. Part of the suggestions from user feedback could not be implemented, so only some were successfully added, such as increased icon visibility, and adjustments to the simulations and notebook. Additionally, the number of participants was increased to 7, in order to get better and more accurate insights. In regards to the questionnaire, this evaluation had a new set of SUS questions for the library hub alone, and the ten SUS questions that were for the whole system, were instead directed to the rest of the system. This decision was made in order to prevent the answers from skewing the overall results in case the library hub performed significantly worse than the rest of it.

These concerns did not come to fruition, as, although the scores decreased slightly, the scores were between 85 and 90 points for both the library hub and the rest of the system. The slight decrease in score can be attributed to the increase in participant count and variability due to a still small sample size, rather than a decrease of quality of the prototype in comparison to the one used in evaluation 1.

Notably, a few participants felt that the system, particularly the library hub, was a bit overly complex. It is understandable, given it contained three different view modes for it, with their own set of interactions, and the prototype allowed users to have access to all functionalities of the website at the same time. However, in the ideal, finished version, all components would be finely interconnected and unlocked as the user progressed, which would alleviate the perceived complexity of the website. Still, it was valuable and valid criticism, so it was kept in mind in view of the final prototype to avoid making it too convoluted. Most of the other suggestions given were successfully implemented for the final prototype, such as page numbering for the book. However, there were some that did not make it to the final prototype, due to a combination of difficulty of implementation and limited time.

For a more detailed look on the evaluation results, refer to the evaluation documents. Overall, the results of both evaluations managed to satisfy our usability specifications:

1. **Effectiveness:** Users demonstrated improved understanding of basic football rules after engaging with the platform.
2. **Clarity:** Interactive visualizations and intuitive design helped simplify complex concepts without oversimplification.
3. **Satisfaction:** Positive feedback highlighted the appeal of the platform's aesthetics and the immersive learning experience.

## 4 Conclusion and Discussion

The evaluations performed indicate that the platform successfully meets its primary goals. Overall, the platform was thought to be a better alternative than traditional football learning resources, was able to teach its users football knowledge, and had a high usability. However, the second evaluation showed that a bigger sample size for the evaluations would provide more accurate insights, as our increase to seven people allowed for some more mixed responses towards the website, which can be more helpful than positive ones in seeing weaknesses of the platform.

All in all, the evaluations proved that the use of different creative interactive information visualizations and the overall book interface in the platform were able to provide an enjoyable and useful experience above traditional means. This directly answers the research question with an affirmative yes.

Despite the positive end result, there were various difficulties and decisions that had to be made throughout the project's creation.

In the initial stages of the project, while writing the project plan, the idea was to have many more functionalities, such as live match visualization in the library hub, and other features such as uploading your own books in the hub. However, the supervisor advised that it might be too ambitious given the limited time available, so most of those additional ideas were cut.

For the evaluations, it was difficult to decide on the questions for the questionnaire for the user evaluations. Due to the website being divided in different components, the ideal outcome would have been to have various questions for each simulation and component of the system. However, that would have resulted in way too many questions, overwhelming the participants. As such, trying to find a balance, a decision was made to have the ten SUS questions for the whole website (and after for the library hub as well) and a series of individual questions for each component and some directly related to the research question.

Furthermore, a significant limiting factor for the project was time. The platform greatly depended on the quality and creativity of the simulations and minigames; there were often moments of stagnation due to a lack of ideas. Having to learn three.js from scratch also meant some restrictions in the scope and complexity of the 3D components. Additional time focusing on learning to use the library would have allowed for higher quality 3D interactions and experiences.

## 5 Future Work

The final prototype could be further expanded upon in multiple ways. The clearest one is the addition of content; although the simulations were filled with some information and a basic story was added for the book, those have great room for improvement. For the sake of the course, the project was done focusing on functionality and interactivity, but in a real setting, the story and information content would be equally as important.

Another exciting possibility links the platform to real-time football matches. The platform could provide real-time explanations of the situation on the field, and other pieces of trivia and information about the game being played. This would enrich the watching experience of users and enhance the learning provided by watching matches.

And lastly, there is also accessibility. A mobile-friendly version would give extra commodity to users, and also allow them to use the platform in places where having a laptop would not be possible, such as standing outdoors. Moreover, adding different languages could open this platform to a much wider audience. These additions could overall raise the degree of professionalism of the project, and make it closer to an actual launch-ready product.

## References

1. Aupperlee, Aaron. 2021. "New Research Shows Learning Is More Effective When Active." Carnegie Mellon University. <https://www.cmu.edu/news/stories/archives/2021/october/active-learning.html?>
2. Buffalo University. n.d. "Evidence of Active Learning's Effectiveness - Office of Curriculum, Assessment and Teaching Transformation." University at Buffalo. <https://www.buffalo.edu/catt/teach/develop/design/designing-activities/active-learning-effectiveness.html?>
3. Kolb, David. 2014. *Experiential Learning: Experience As the Source of Learning and Development*. N.p.: Pearson Education, Limited. <https://books.google.es/books?hl=es&lr=&id=jpbeBQAAQBAJ&oi=fnd&pg=PR7&dq=experiential+learning&ots=Vp5TkQXVMa&sig=OYO4LZnfsWR249tpBF0ZBvGS4Os#v=onepage&q=experiential%20learning&f=false>
4. O'Hanlon, Ryan. 2022. "How to watch soccer smarter: What to watch beyond the ball itself." ESPN. [https://www.espn.com/sports/soccer/insider/story/\\_/id/34894673/how-watch-understand-soccer-following-ball](https://www.espn.com/sports/soccer/insider/story/_/id/34894673/how-watch-understand-soccer-following-ball)
5. Soccer.com. 2022. "The Basic Rules of Soccer: A Complete Guide | SOCCER.COM." Soccer.com. <https://www.soccer.com/guide/rules-of-soccer-guide>
6. US National Soccer Players. n.d. "Beginners Guide to Soccer." US Soccer Players. <https://ussoccerplayers.com/beginners-guide-to-soccer?>
7. WorldAtlas. 2023. "The Most Popular Sports In The World - WorldAtlas." World Atlas.

[https://www.worldatlas.com/articles/what-are-the-most-popular-sports-in-the-world.h  
tml](https://www.worldatlas.com/articles/what-are-the-most-popular-sports-in-the-world.html)

8. World Soccer Talk. n.d. "Beginner's Guide To Soccer Rules." World Soccer Talk. <https://worldsoccertalk.com/beginners-guide-to-soccer-rules/>

## Appendix

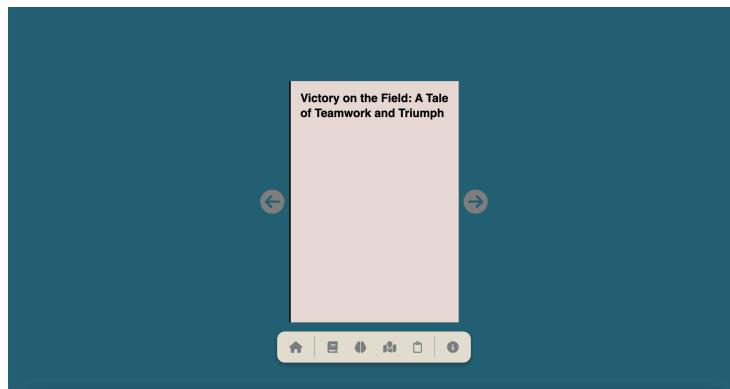


Figure 1: Introductory book



Figure 2: Formations Simulation

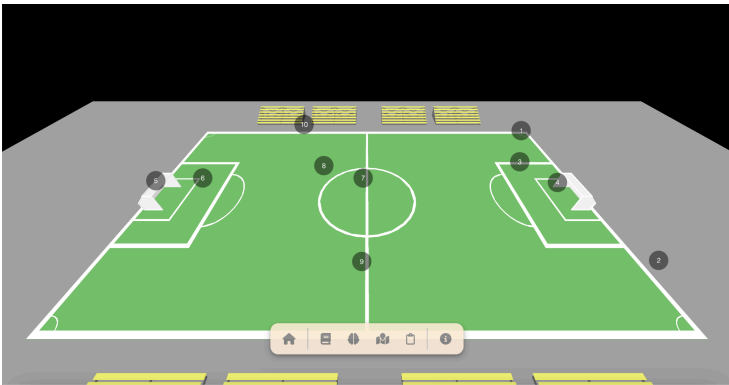


Figure 3: Field Simulator

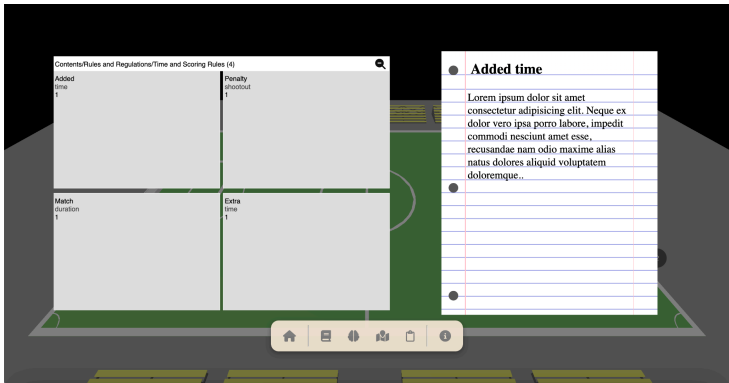


Figure 4: Definitions Guidebook



Figure 5: Traditional Library View



Figure 6: Interactive Library View

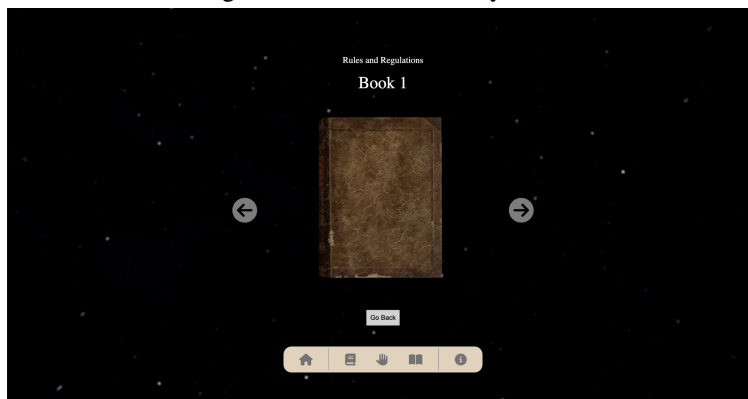


Figure 7: Interactive Library View after clicking on a section

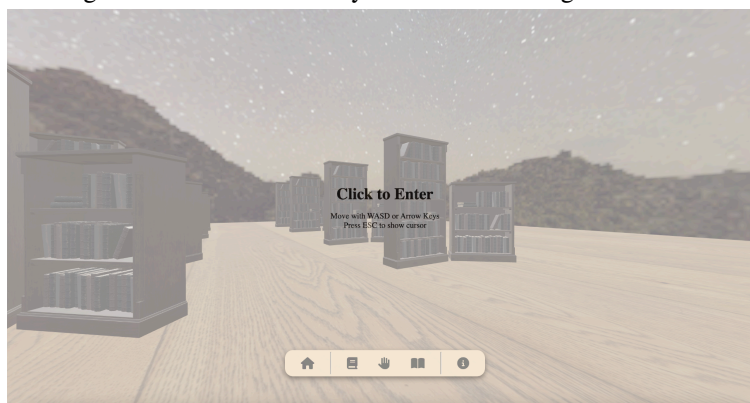


Figure 8: Immersive Library View